



PRODUCT
DATASHEET '24

ADVANCE ANALYTIK

REVOLUTIONIZING ONLINE MONITORING SOLUTIONS



OPTICS 1000 – Total Chromium

TOTAL CHROMIUM (Cr)

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Optics 1000 Series

TOTAL CHROMIUM (Cr)

Method - Colorimetric

After adding the sample into the measurement cell, some reagents are added in order to adjust the solution to the desired conditions (pH, valence's elements, etc.). Then, a blank is done to correct any temperature or turbidity disturbance. Subsequently, a last reagent is added, and it reacts with solution developing a color, which is measured using a correct wavelength. Thanks to the photometer used, the result achieves a great accuracy.



Principle of measurement

Cerium sulfate is added, which oxidizes the sample and converts all forms of chromium to chromium VI. Then diphenylcarbohydrazide is added which reacts with the chromium VI to produce a reddish-purple color.



Advantages of the method

The method is very simple, very specific and is very sensitive because of the high absorbance of the diphenylcarbazide-chromium(VI) complex formed. Chromium (III) can be measuring calculating the difference between Cr Total and Cr (VI).

Specifications

RANGE	From 0 to 200 ppb. Adjustable higher concentrations with internal dilution.
ACCURACY	±2%
REPEATABILITY	±2%
ANALYSIS TIME	around 10 minutes
CALIBRATION	two point
LED WAVELENGTH	545 nm

Reagents consumption

- Reagent 1: 2.5 ml / analysis - 2.0L / month
- Reagent 2: 0.8 ml / analysis - 0.75L / month

Monthly consumption calculated assuming 1 analysis per hour.



**Note -**

This data sheet serves as general information about the Optics 1000 - Total Chromium (Cr). For specific technical details, installation guidelines, and troubleshooting assistance, please refer to the official user manual provided with the product.

For inquiries and detailed technical information, please contact sales@advanceanalytik.com.



LET'S CONNECT !

Let's work together to find a solution that works for you



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